



DO-IT-YOURSELF: PROJECT



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ously scans a half-circle area (0-180°) within a radius of 500cm (the maximum range of the HC-SR04 sensor). When it detects an object at any angle, it immediately measures its distance and transmits both data to a remote receiver located more than a kilometre away. The receiver's TFT display acts as the radar monitor, showing the distance and angle of detected objects with multi-colour graphics and animation.

Software

The circuit operation is controlled by software programs uploaded into the internal flash memory of both transmitter and receiver Arduino boards (ATMega328). There are two programs: one for the transmitter and one for the receiver. The programs are written in Arduino C/C++ language, compiled and built using the Arduino IDE software, and uploaded into the flash memory of the ATMega328.

The transmitter program performs the following tasks:

1. It rotates the servo motor from 0-180° and back to 0° by generating a PWM signal on analogue output pin D11
2. It detects any object in front of it using the HC-SR04 module and measures its distance
3. It reads the servo motor angle when an object is detected
4. It serially transmits this distance and angle data using the HC-12 module

It uses the following libraries:

1. **Servo**. Used to rotate the servo motor to the desired angle, clockwise and anti-clockwise
2. **SoftwareSerial**. Used to communicate with the HC-12 module. It sends data serially using D4 (UART Tx) pin to the HC-12. This library allows D3 and D4 pins of the Arduino to function as Rx and Tx pins for UART

The receiver program performs

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The source code of this project is available for free download at www.electronicsforu.com

the following tasks:

1. It displays the radar scanning process animation on the TFT monitor
2. It serially receives data from the HC-12 module
3. It displays the distance and angle of the detected object on the TFT monitor

It uses the following libraries:

1. **LCDWIKI_GUI.h**. Used to display characters, numbers, shapes, animations, etc, and create a GUI on the TFT display
2. **LCDWIKI_SPI.h**. Used to communicate with the SPI-type TFT display using a 4-wire SPI protocol
3. **SoftwareSerial**. Used to communicate with the HC-12 module, similar to the transmitter side. It receives data serially using D3 (UART Rx) pin from the HC-12. This library allows D3 and D4 pins of the Arduino to function as Rx and Tx pins for UART

Construction and testing

For the construction of the entire device, refer to the wiring diagrams of the transmitter (Fig. 6) and the receiver (Fig. 7). Also, refer to Fig. 8 before constructing the project. Do not forget to upload the source codes into the transmitter and receiver Arduino Nano boards before assembling the circuit.

After properly assembling both the transmitter and receiver circuits, power both devices, and you will see the output on the display on the receiver side. **EFY**

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